

Op-Amp-Based VEHICLE THEFT DETECTOR

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A simple, low-cost device can effectively alert homeowners or occupants if a parked

vehicle is moved or tampered with. This hardwired detector features a main circuit with a buzzer alarm

installed inside the house, while a sensor is mounted on the vehicle being protected.

The sensor should be positioned on the vehicle in a location shielded from heat, rainwater, dust, and dirt. Its leads must be routed to an accessible point outside the vehicle for easy connection to the wires coming from the main circuit.

For consistent protection, the sensor leads must remain connected to the main circuit whenever the vehicle is parked. Any disconnection, cutting, or shorting of the wires triggers the alarm and activates the LED indicator. The detector is powered by an external 12V DC adaptor, with a 9V dry cell providing backup during power failures. Fig. 1 displays the author's prototype of the detector.

Circuit and working

Fig. 2 shows the circuit diagram of the vehicle theft detector. The system



Fig. 1: Author's prototype

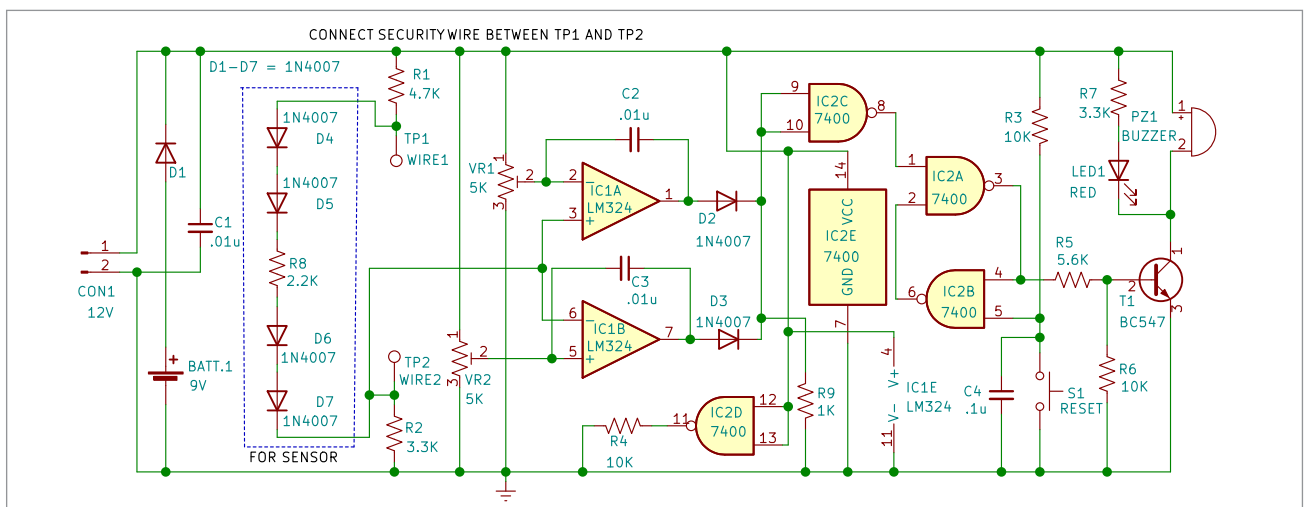


Fig. 2: Circuit diagram